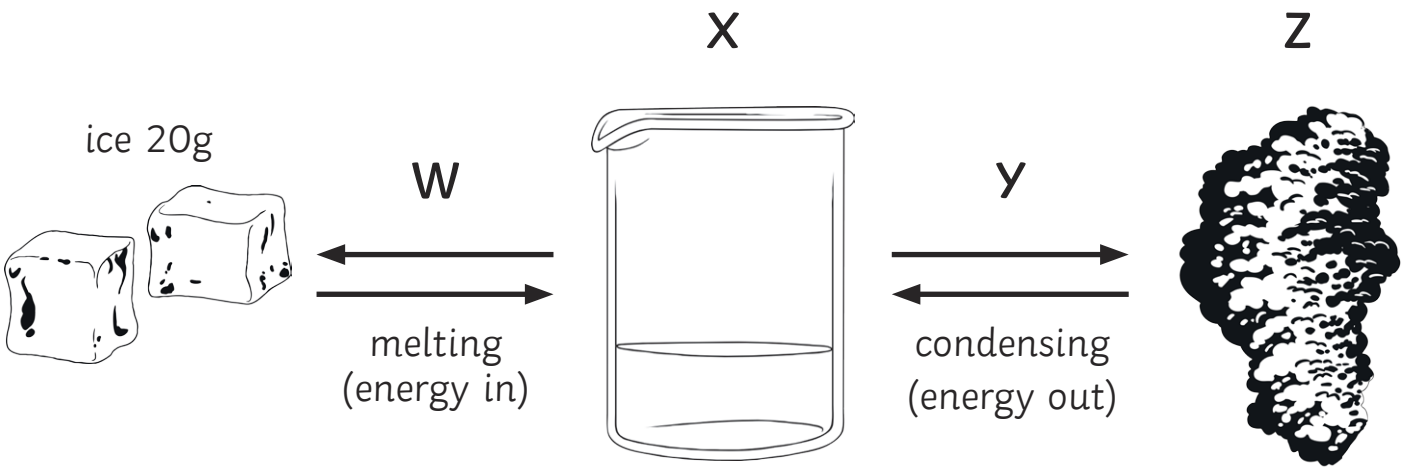
Name:	Class:	Date given:
		Date due in:

Energy in Matter

Look at the diagram below. It is not completely labelled. Answer the questions below about the missing labels.



1a. What is X? [1]

1b. What is the mass of X? [1]

1c. What is Z? [1]

1d. What is the mass of Z? [1]

1e. What process is Y? [1]

1f. Is energy in or out at Y? [1]

1g. What process is W? [1]

1h. Is energy in or out at W? [1]

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2a. If energy is gained during a reaction is it: (tick one) [1]

- ☐ Endothermic?
- ☐ Exothermic?

2b. If energy is gained during a reaction is it: (tick one) [1]

- ☐ Endothermic?
- ☐ Exothermic?

2c. Where is the 'lost' energy transferred? [1]

_____

3. Explain, in terms of particles, why a gain in energy can make a liquid change state into a gas. [2]

_____

_____

Learning Outcomes (tick if achieved)

Q1	I understanding changes of state require energy	
Q2	I can describe types of reactions in terms of energy transfer	
Q3	I can describe change of state in terms of energy	

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Energy in Matter **Answers**

1a. What is X? [1]

water (liquid)

1b. What is the mass of X? [1]

20g

1c. What is Z? [1]

steam/vapour (gas)

1d. What is the mass of Z? [1]

20g

1e. What process is Y? [1]

evaporating

1f. Is energy in or out at Y? [1]

in

1g. What process is W? [1]

freezing

1h. Is energy in or out at W? [1]

out

2a. If energy is gained during a reaction is it: (tick one) [1]

endothermic

2b. If energy is lost during a reaction is it: (tick one) [1]

exothermic

2c. Where is the 'lost' energy transferred? [1]

to the surroundings

3. Explain, in terms of particles, why a gain in energy can make a liquid change state into a gas. [2]

- **particles gain energy and move around more**
- **spread out more and becomes a gas substance**